3 Regional or dialectal variation in Bangla

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Acoustic Analysis of the Speakers' Variability for Regional Accent-Affected Pronunciation in Bangladeshi Bangla: A Study on Sylheti Accent

Title: Acoustic Analysis of the Speakers' Variability for Regional Accent-Affected Pronunciation in Bangladeshi Bangla: A Study on Sylheti Accent

Year: 2020

BibTeX Citation:

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@article{article,
author = {Kibria, Shafkat and Rahman, M. and Selim, Mohammad and Iqbal, Muhammed},
year = {2020},
month = {02},
pages = {35200-35221},
title = {Acoustic Analysis of the Speakers' Variability for Regional Accent-Affected Pronunciation in Bangladeshi Bangla: A Study on Sylheti Accent},
volume = {8},
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doi = {10.1109/ACCESS.2020.2974799}
}
```

Which regional variety/varieties:

Sylhet region (extremely deviant dialect) compared against a "neutral" baseline from the North-West and middle parts of Bangladesh.

Method type:

Quantitative acoustic-phonetic analysis and machine learning classification.

Phonetic feature studied – vowels / consonants / sibilants / tone-pitch / rhythm / duration:

Vowels (specifically formant frequencies F1, F2, F3, pitch slope, and vowel duration). It explicitly excludes consonants from the analysis.

ABSTRACT Accented pronunciation variability is one of the key elements that deteriorate the accuracy of the automatic speech recognition (ASR). This article reports the results of the acoustic analysis of the two groups of speakers' variability caused by regional accent in Bangladeshi Bangla. The analysis considers the seven monophthongal and four diphthongal vowels of Bangla to investigate the acoustic characteristics of two groups of single-accent speakers and their correlation on the articulation of the Standard Colloquial Bangladeshi Bangla (SCBB). An accent is the speaker's regional signature and shaped by his/her community and educational background. This study examines both male and female speakers from the Sylhet region, which has one of the extremely deviant dialects in Bangla, and comparatively less deviant speakers from different districts of North-West and Middle Part of Bangladesh. Accent-related acoustic features such as pitch slope, formant frequencies, and vowel duration have been considered to examine the prominent

Data used – how many speakers, read or spontaneous, parallel or not?:

15 speakers in total (7 Sylheti, 8 Neutral) utilizing a parallel, read-speech design.

Key finding:

Significant acoustic deviations exist between the Sylheti dialect and standard Bangla regarding vowel formants and pitch steepness, and these specific deviations measurably degrade the performance of standard Automatic Speech Recognition (ASR) systems.

Strong or subtle/qualitative?

Strong/Quantitative. The differences are measured acoustically and proven to be statistically significant using rigorous t-tests.

The /s/→/ʃ/ or Sylhet-sibilant claim – if present, capture the exact claim and whether it was measured or asserted:

This claim is not present in this paper. The authors explicitly state that they chose to restrict their study entirely to vowels because vowels carry more acoustic detail related to accents than consonants do.

Dataset on multiregional variations of Bangla language (BD-Dialect)

Title: Dataset on multiregional variations of Bangla language (BD-Dialect)

Year: 2026

BibTeX Citation:

@article{article,
author = {Rahman, Anika and Muna, Nafesha and Prity, Masuma},
year = {2026},
month = {03},
pages = {112654},
title = {Dataset on Multiregional Variations of Bangla Language (BD-Dialect)},
volume = {66},
journal = {Data in Brief},
doi = {10.1016/j.dib.2026.112654}

}

Which regional variety/varieties:

Standard Bangla alongside five major regional dialects: Noakhali, Sylhet, Chittagong, Rajshahi, and Mymensingh.

Method type:

Text-based / Computational (Corpus Creation). The study focuses on building a textual dataset through translation, transcription, and manual validation rather than acoustic-phonetic measurement.

Phonetic feature studied – vowels / consonants / sibilants / tone-pitch / rhythm / duration:

None. The paper focuses entirely on lexical (vocabulary) variation and text translation rather than phonetic or acoustic analysis.

Data used – how many speakers, read or spontaneous, parallel or not?:

Speakers: 19 total native speakers (Noakhali: 3, Sylhet: 3, Chittagong: 3, Rajshahi: 5, Mymensingh: 5) were used for data elicitation and validation.

Style: Mostly elicited text translations and regional literature rather than natural spoken speech. A very small subset of pilot audio was collected but was not the primary data.

Parallel: Yes, highly parallel (aligned translations of the exact same words/clauses across all regions).

Table 3
Native speaker participant summary.

Dialect Region	Number of Speakers	Age Range	Gender (M/F)	Primary Role
Noakhali	3	22–30	3 / 0	Data Elicitation & Validation
Sylhet	3	20–30	0 / 3	Data Elicitation & Validation
Chittagong	3	21–32	0 / 3	Data Elicitation & Validation
Rajshahi	5	21–28	3 / 2	Data Elicitation & Validation
Mymensingh	5	23–30	2 / 3	Data Elicitation & Validation

Key finding:

The authors successfully constructed and validated an open-access, parallel textual corpus of 950 words and clauses across five major Bangla dialects to aid natural language processing and cross-linguistic studies.

Strong or subtle/qualitative?: Qualitative/Textual. The research output is a categorical translation database rather than a measurement of subtle acoustic shifts.

The /s/→/ʃ/ or Sylhet-sibilant claim – if present, capture the exact claim and whether it was measured or asserted:

This claim is not present in this paper. The dataset documents textual and lexical translation alignment, not phonetic or sibilant realization tracking.

RegSpeech12: A Regional Corpus of Bengali Spontaneous Speech Across Dialects

Title: RegSpeech12: A Regional Corpus of Bengali Spontaneous Speech Across Dialects

Year: 2025

BibTeX Citation:

```
@unknown{unknown,
author = {Hassan, Md and Hossain, Azmol and Fatema, Kanij and
Faruque, Rubayet and Shome, Tanmoy and Naswan, Ruwad and
Chakraborty, Trina and Zihad, Md and Dipto, Tawsif and Tasnim,
Nazia and Ansary, Nazmuddoha and Shawon, Md Mehedi Hasan and
Humayun, Ahmed and Alam, Md Golam Rabiul and Sadeque, Farig and
Sushmit, Asif},
year = {2025},
month = {10},
pages = {},
title = {RegSpeech12: A Regional Corpus of Bengali Spontaneous
Speech Across Dialects},
doi = {10.48550/arXiv.2510.24096}
}
```

Which regional variety/varieties: 12 distinct regional dialects across Bangladesh.

Method type: Computational (Spontaneous Corpus Creation). The researchers built an unprompted speech dataset to test ASR (Automatic Speech Recognition) models rather than conducting targeted phonetic experiments.

Phonetic feature studied – vowels / consonants / sibilants / tone-pitch / rhythm / duration:

General spectral and prosodic variations using the Long-Term Average Spectrum (LTAS). The study does not isolate specific vowels, consonants, or sibilants.

Data used – how many speakers, read or spontaneous, parallel or

not?:

394 speakers utilizing spontaneous (unprompted), non-parallel speech.

- *Gender diversification:* In constructing the speech corpus, we placed emphasis on achieving a balanced distribution of genders, aiming for a 50:50 ratio between male and female voices. The final collection consists of contributions from at least 394 speakers. Of these, approximately 53.04% are male and 37.56% are female, while 9.39% of the recordings include mixed-gender interactions with multiple speakers. This corresponds to 209 male participants, 148 female participants, and 37 clips featuring both genders. The overall gender distribution across districts is illustrated in Fig. 2.

Key finding: The authors successfully created the first publicly available, large-scale ASR dataset focused on spontaneous regional Bangla dialects. They found that while these dialects have massive vocabulary differences, their acoustic profiles actually overlap significantly.

Strong or subtle/qualitative?:

Subtle acoustic divergence. Unlike paper #13 which found strong acoustic differences in isolated vowels, this spontaneous speech study found that overall acoustic clustering between dialects is weak.

The /s/→/ʃ/ or Sylhet-sibilant claim — if present, capture the exact claim and whether it was measured or asserted:

thos claim is not present

Phonological variation and linguistic diversity in Bangladeshi dialects: An exploration of sound patterns and sociolinguistic significance

Title: Phonological variation and linguistic diversity in Bangladeshi dialects: An exploration of sound patterns and sociolinguistic significance

Year: 2024

BibTeX Citation:

@article{article,
author = {Rahman, Mohammad and Barman, Binoy and Sharmin, and Uddin, Md and Yusuf, Sakiba and Rasool, Ushba},
year = {2024},
month = {05},
pages = {1188},
title = {Phonological variation and linguistic diversity in Bangladeshi dialects: An exploration of sound patterns and sociolinguistic significance},
volume = {6},
journal = {Forum for Linguistic Studies},
doi = {10.59400/fls.v6i2.1188}
}

Which regional variety/varieties: 20 distinct regions across Bangladesh.

Method type: Descriptive, qualitative, and transcription-based.

Phonetic feature studied – vowels / consonants / sibilants / tone-pitch / rhythm / duration:

Consonants (specifically focusing on categorical changes in plosives, fricatives, affricates, and shifts in aspiration or voicing).

Data used – how many speakers, read or spontaneous, parallel or not?:

70 speakers using elicited/parallel speech (a standardized set of target words).

3.3. Procedure

The procedure involved a meticulous approach to collect and analyze phonological variations across 20 regions of Bangladesh. Semi-structured interviews were conducted with 70 participants, selected from diverse regions and fluent in their local dialects. A structured word list and detailed interview guide were utilized to elicit specific phonological patterns. Data collection was followed by immediate

Key finding:

The study identified 13 distinct and systematic phonological pattern changes across Bangladeshi dialects.

Abstract: This paper explores how people in different parts of Bangladesh talk differently from each other. The study pays special attention to the unique sounds in their dialects compared to the standard Bangla language. The main goal of the research is to understand why these sounds change from one dialect to another. The researchers collected data from 70 participants coming from 20 regions of Bangladesh, who were fluent in their regional dialects and studied it carefully. By employing semi-structured interviews, IPA transcription, and thorough verification processes, this study ensures the accuracy and reliability of its findings and discussion, contributing valuable insights into the phonological diversity of Bangladeshi dialects and their socio-linguistic significance. The authors found 13 phonological variations or patterns among the dialects of different regions of Bangladesh. Each type of change is

Abstract: This paper explores how people in different parts of Bangladesh talk differently from each other. The study pays special attention to the unique sounds in their dialects compared to the standard Bangla language. The main goal of the research is to understand why these sounds change from one dialect to another. The researchers collected data from 70 participants coming from 20 regions of Bangladesh, who were fluent in their regional dialects and studied it carefully. By employing semi-structured interviews, IPA transcription, and thorough verification processes, this study ensures the accuracy and reliability of its findings and discussion, contributing valuable insights into the phonological diversity of Bangladeshi dialects and their socio-linguistic significance. The authors found 13 phonological variations or patterns among the dialects of different regions of Bangladesh. Each type of change is explained with clear examples to make it easier to understand. The research contributes significantly to the understanding of socio-linguistic diversity in Bangladesh by providing experimental evidence of phonological variation and its determinants. It also lays the groundwork for future research on dialects and their relationship with the standard language. This study will bring up a greater appreciation for Bangladesh's linguistic diversity and encourage further research into language variation and change.

Strong or subtle/qualitative?:

Qualitative. The study presents these variations as categorical, absolute phonetic shifts based on transcription rather than subtle acoustic measurements.

The /s/→/ʃ/ or Sylhet-sibilant claim — if present, capture the exact claim and whether it was measured or asserted:

The claim is present and is asserted through IPA transcription rather than acoustically measured. The paper documents a categorical shift from the palatal fricative /ʃ/ to the alveolar fricative /s/ (and also affricates to /s/).

Pattern 3:

Palatal fricative to alveolar fricative: /ʃ/ → /s/ শ → স

In this pattern /ʃ/ becomes /s/ in people's usual pronunciation in Chapainababganj, Rajshahi, Dinajpur, Old Dhaka and adjacent areas. Here a

Pattern 5:

Unaspirated affricate to alveolar fricative: /c/ → /s/ চ → স

In this pattern an affricate sound becomes an unaspirated alveolar fricative, in the pronunciation of the people in Narayanganj, Comilla, Brahmanbaria, Mymensingh and adjacent areas (see **Table 5**).

Table 5. Unaspirated affricate to alveolar fricative.

Word in standard Bangla	Phonetic transcription	Word in dialect	Phonetic transcription	Meaning of word
pocha	/pɒca/	posa	/pɒsa/	Rotten
kach	/kac/	kas	/kas/	Glass
kancha	/kaca/	kasa	/kasa/	Raw
moch	/moc	mos	/mos/	Moustache
bancha	/banca/	basa	/basa/	To live
kochi	/kɒci/	kosi	/kɒsi/	Tender/green
khancha	/kʰanca/	khasa	/kʰasa/	Cage
khoncha	/kʰonca/	khosa	/kʰosa/	Prick
panch	/panc/	pas	/pas/	Five
kochu	/kɒcu/	kosu	/kɒsu/	Arum

Table 5. Unaspirated affricate to alveolar fricative.

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kach	/kac/	kas	/kas/	Glass
kancha	/kaca/	kasa	/kasa/	Raw
moch	/moc	mos	/mos/	Moustache
bancha	/banca/	basa	/basa/	To live
kochi	/kɒci/	kosi	/kɒsi/	Tender/green
khancha	/k^{h}anca/	khasa	/k^{h}asa/	Cage
khoncha	/k^{h}onca/	khosa	/k^{h}osa/	Prick
panch	/panc/	pas	/pas/	Five
kochu	/kɒcu/	kosu	/kɒsu/	Arum
nich	/nic/	nis	/nis/	Mean/low